

## 2501.06662: equation evidence

### Display equations

| Identifier             | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                               | Rendered                                                                                                                                                                                                                                                                                                                  | Image                                                                                                                                                                                                                                                                                                                     |
|------------------------|------|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2501.06662_E00051      | 21   | ■ 0.536 | $\begin{aligned} \operatorname{Mag}(t \mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t. \end{aligned}$ | $\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t.$ | $\operatorname{Mag}(t \mathcal{M}) = \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y)$ $= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t.$ |
| 2501.06662_E00059      | 23   | ■ 0.550 | $\begin{aligned} \lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n} &= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1}   \perp) \log \pi(\perp a_1 \cdots a_{N-1}   \perp) \end{aligned}$                                                      | $\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1}   \perp) \log \pi(\perp a_1 \cdots a_{N-1}   \perp)$                                                  | $\lim_{n \rightarrow \infty} \frac{\mathcal{H}(\mathcal{M}, \pi(- \perp) _{T(\perp)}, \zeta)}{n}$ $= \lim_{n \rightarrow \infty} -\frac{1}{n} \sum_{(a_1, \dots, a_{N-1}) \in A^{N-1}} \pi(\perp a_1 \cdots a_{N-1}   \perp) \log \pi(\perp a_1 \cdots a_{N-1}   \perp)$                                                  |
| 2501.06662_E00058      | 23   | ■ 0.594 | $\begin{aligned} \mathcal{H}(\mathcal{M}, p, \zeta) &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right) \\ &= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z   y)^t p(z) \right). \end{aligned}$    | $\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z   y)^t p(z) \right).$    | $\mathcal{H}(\mathcal{M}, p, \zeta) = - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \zeta_t(y, z) p(z) \right)$ $= - \sum_{y \in \operatorname{ob}(\mathcal{M})} p(y) \log \left( \sum_{z \in \operatorname{ob}(\mathcal{M})} \pi(z   y)^t p(z) \right).$    |
| 2501.06662_E00023      | 15   | ■ 0.636 | $\begin{aligned} \#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} &= \sum_{i=0}^{N-2} \#A^i \\ &= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1. \end{aligned}$                                                                                                                             | $\#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i\right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1.$                                                                                            | $\#\left\{x \in \operatorname{ob}(\mathcal{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i\right\} (\#A + 1) = \left(\sum_{i=0}^{N-2} \#A^i\right) (\#A + 1)$ $= \#A^{N-1} + 2(\#A^{N-2}) + \dots + 2(\#A) + 1.$                                                                                            |
| 2501.06662_E00052 (13) | 21   | ■ 0.693 | $\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$                                                                                                                                                                                                                              | $\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$                                                                                                                                                                                             | $\operatorname{Mag}(t \mathcal{M}) = \sum_{k \geq 0} \sum_{\ell \in [0, \infty)} \sum_{c \in G_{k, \ell}} (-1)^k e^{-t\ell}.$                                                                                                                                                                                             |

| Identifier             | Page | Conf.   | LaTeX source                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Rendered                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Image                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
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| 2501.06662_E00033      | 17   | ■ 0.720 | $\zeta_{\mathbf{t}}^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^{\mathbf{t}} & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                                                           | $\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                                                                      | $\zeta_t^{-1}(x, y) = \begin{cases} -\pi(y \mid x)^t & \text{if } y \in \mathcal{L}_x^{(1)} \\ 1 & \text{if } y = x \\ 0 & \text{otherwise.} \end{cases}$                                                                                                                                                                                                                                                                                                                     |
| 2501.06662_E00037 (10) | 18   | ■ 0.780 | $\begin{aligned} \operatorname{Mag}(\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_{\mathbf{t}}^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_{\mathbf{t}}^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^{\mathbf{t}} \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left( 1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^{\mathbf{t}} \right) + \#(T(\perp)). \end{aligned}$ | $\begin{aligned} \operatorname{Mag}(t\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left( 1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$ | $\begin{aligned} \operatorname{Mag}(t\mathcal{M}) &= \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, x) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \sum_{y \in \mathcal{L}_x^{(1)}} \pi(y \mid x)^t \\ &= \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} \left( 1 - \sum_{a \in A \cup \{\dagger\}} p_x(a)^t \right) + \#(T(\perp)). \end{aligned}$ |
| 2501.06662_E00049      | 20   | ■ 0.797 | $\begin{aligned} \partial_k^i(y_0, \dots, y_k) &= \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases} \\ \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) & \\ \text{otherwise} & \end{aligned}$                                                                                                                                                                                         | $\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                           | $\partial_k^i(y_0, \dots, y_k) = \begin{cases} (y_0, \dots, y_{i-1}, y_{i+1}, \dots, y_k) & \text{if } d(y_{i-1}, y_i) + d(y_i, y_{i+1}) = d(y_{i-1}, y_{i+1}) \\ 0 & \text{otherwise} \end{cases}$                                                                                                                                                                                                                                                                           |
| 2501.06662_E00001      | 3    | ■ 0.822 | $\operatorname{Mag}(\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                                                                           | $\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                               | $\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                               |
| 2501.06662_E00029      | 16   | ■ 0.829 | $\left( \zeta_{\mathbf{t}} - \delta \right)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^{\mathbf{t}}.$                                                                                                                                                                                                                                                                                        | $(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$                                                                                                                                                                                                                                                                                | $(\zeta_t - \delta)^k(x, y) = \sum_{\substack{(y_0, y_1, \dots, y_{k-1}, y_k) \\ y_0 = x, y_k = y \\ (y_{i-1}, y_i) \in E \text{ for } i=1, \dots, k}} \prod_{i=1}^k \pi(y_i \mid y_{i-1})^t.$                                                                                                                                                                                                                                                                                |
| 2501.06662_E00036 (11) | 18   | ■ 0.835 | $\operatorname{Mag}(\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                                                                           | $\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                               | $\operatorname{Mag}(t\mathcal{M}) = (t-1) \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H_t(p_x) + \#(T(\perp)), \quad t > 0.$                                                                                                                                                                                                                                                                                                                               |

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| 2501.06662_E00026 (8)  | 15   | ■ 0.842 | $\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t.$                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t.$                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\zeta_t^{-1}(x, y) = \sum_{k \geq 0} \sum_{\substack{\text{nondeg. paths} \\ x=y_0 \rightarrow y_1 \rightarrow \dots \rightarrow y_k=y}} (-1)^k \prod_{i=1}^k \pi(y_i   y_{i-1})^t.$                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2501.06662_E00021      | 14   | ■ 0.849 | $\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1). \end{aligned}$ | $\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1) \end{aligned}$ | $\begin{aligned} \operatorname{Mag}(\mathrm{L}) &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \zeta_{\mathrm{L}}^{-1}(x, y) \\ &= \sum_{x, y \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, y) \\ &= \sum_{x \in \operatorname{ob}(\mathrm{L})} \mu_{\mathrm{L}}(x, x) + \sum_{\substack{x=\perp a \in \operatorname{ob}(\mathrm{L}) \\ a \in \bigcup_{i=0}^{N-2} A^i}} \sum_{a_1 \in A \cup \{\dagger\}} \mu_{\mathrm{L}}(x, xa_1) \\ &= \# \operatorname{ob}(\mathrm{L}) - \# \left\{ x \in \operatorname{ob}(\mathrm{L}) : x = \perp a \text{ with } a \in \bigcup_{i=0}^{N-2} A^i \right\} (\#A + 1). \end{aligned}$ |
| 2501.06662_E00022      | 15   | ■ 0.871 | $\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2 \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                     | $\begin{aligned} \# \operatorname{ob}(\mathrm{L}) &= 1 + \sum_{i=1}^{N-1} \# \mathrm{L}^{(i)} \\ &= 1 + \sum_{i=1}^{N-1} \# A^i + \# A^{i-1} = \# A^{N-1} + 2(\# A^{N-2}) + \dots + 2(\# A) + 2. \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2501.06662_E00045      | 19   | ■ 0.922 | $\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | $\zeta_{\mathcal{M}_x}^{-1} = \zeta_{\mathcal{M}}^{-1} \big _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)},$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| 2501.06662_E00039 (12) | 18   | ■ 0.940 | $f'(\text{\textbackslash prime})(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | $f'(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | $f'(1) = \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} H(p_x).$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

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| 2501.06662_E00046     | 20   | ■ 0.942 | $\begin{aligned} \operatorname{Mag}(\mathcal{M}_x) &= \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ &= \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ &= (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)). \end{aligned}$ | $\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$ | $\operatorname{Mag}(t\mathcal{M}_x) = \sum_{y, z \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, z) \\ = \sum_{y \in \operatorname{ob}(\mathcal{M}_x)} \zeta_{\mathcal{M}_x}^{-1}(y, y) - \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} \sum_{y \in L_y^{(1)}} \pi(z y)^t \\ = (t-1) \sum_{y \in \operatorname{ob}(\mathcal{M}_x) \setminus T(x)} H_t(p_y) + \#(T(x)).$ |
| 2501.06662_E00040     | 19   | ■ 0.945 | $f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$                                                                                                                                                                                                                                                                                                                       | $f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$                                                                                                                                                                                                                                                                                       | $f(t) = \#(\operatorname{ob}(\mathcal{M})) - \sum_{x \in \operatorname{ob}(\mathcal{M}) \setminus T(\perp)} Z_x(t),$                                                                                                                                                                                                                                                                                       |
| 2501.06662_E00044     | 19   | ■ 0.971 | $\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$                                                                                                                                                                                                                                                                                                                  | $\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$                                                                                                                                                                                                                                                                                  | $\zeta_{\mathcal{M}_x} = \zeta_{\mathcal{M}} _{\operatorname{ob}(\mathcal{M}_x) \times \operatorname{ob}(\mathcal{M}_x)}$                                                                                                                                                                                                                                                                                  |
| 2501.06662_E00032     | 17   | ■ 0.973 | $\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$                                                                                                                                                                                                                                                                                                 | $\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$                                                                                                                                                                                                                                                                 | $\pi(y x)^t \sum_{k \geq 0} (-1)^k \# \{ \text{nondegenerate paths of length } k \text{ from } x \text{ to } y \text{ in } \mathcal{L} \}$                                                                                                                                                                                                                                                                 |
| 2501.06662_E00056     | 22   | ■ 0.986 | $PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$                                                                                                                                                                                                                                                                                                                                                                                     | $PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$                                                                                                                                                                                                                                                                                                                                                     | $PPL(y) = 1/\zeta_t(a_0, y) = 1/e^{t \ln \pi(y a_0)},$                                                                                                                                                                                                                                                                                                                                                     |
| 2501.06662_E00015 (7) | 12   | ■ 0.988 | $\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$                                                                                                                                                                                                                                                                                 | $\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$                                                                                                                                                                                                                                                 | $\operatorname{Mag}(\mathcal{C}) = \sum_{(x, y) \in \operatorname{ob}(\mathcal{C}) \times \operatorname{ob}(\mathcal{C})} \zeta_{\mathcal{C}}^{-1}(x, y).$                                                                                                                                                                                                                                                 |
| 2501.06662_E00002     | 4    | ■ 0.991 | $\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$                                                                                                                                                                                                                                                                                                                                  | $\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$                                                                                                                                                                                                                                                                                                  | $\frac{(t-1)(n^{1-t}-1)}{1-t} \cdot \#(\operatorname{ob}(\mathcal{M}) \setminus T(\perp)) + \#(T(\perp))$                                                                                                                                                                                                                                                                                                  |
| 2501.06662_E00005     | 6    | ■ 0.991 | $\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and }  a  \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and }  a  < N-1 \}.$                                                                                                                                                                                                                                                                                    | $\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and }  a  \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and }  a  < N-1 \}.$                                                                                                                                                                                                                                                    | $\operatorname{ob}(\mathcal{L}) = \{ \perp a : a \in A^* \text{ and }  a  \leq N-1 \} \sqcup \{ \perp a^\dagger : a \in A^* \text{ and }  a  < N-1 \}.$                                                                                                                                                                                                                                                    |
| 2501.06662_E00006     | 8    | ■ 0.996 | $\begin{aligned} T(x) &= \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ &\quad y \text{ is a finished text such that }  y  \leq N \}. \end{aligned}$                                                                                                                                                                                                                               | $T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that }  y  \leq N \}.$                                                                                                                                                                                                                                     | $T(x) = \{ y \in \operatorname{ob}(\mathcal{L}_x) : y \text{ is an unfinished text of length } N \text{ or} \\ y \text{ is a finished text such that }  y  \leq N \}.$                                                                                                                                                                                                                                     |
| 2501.06662_E00034     | 17   | ■ 0.998 | $\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$                                                                                                                                                                                                                                                                                                                                   | $\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$                                                                                                                                                                                                                                                                                                   | $\operatorname{Mag}(t\mathcal{M}) := \sum_{x, y \in \operatorname{ob}(\mathcal{M})} \zeta_t^{-1}(x, y).$                                                                                                                                                                                                                                                                                                   |

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| 2501.06662_E00003      | 4    | ■ 0.998 | $\underbrace{\#(T(\perp))}_{\leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t\mathcal{M})} \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\operatorname{Mag}(t\mathcal{M}) \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t\mathcal{M})}$ . | $\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t\mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}$ . | $\underbrace{\#(T(\perp))}_{\text{deterministic LM}} \leq \lim_{t \rightarrow \infty} \operatorname{Mag}(t\mathcal{M}) \leq \underbrace{\#(\operatorname{ob}(\mathcal{M}))}_{\text{maximally random LM}}$ . |
| 2501.06662_E00054 (14) | 22   | ■ 0.998 | $\operatorname{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M})$ .                                                                                                     | $\operatorname{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M})$ .                                                     | $\operatorname{Mag}(t\mathcal{M}) = \operatorname{rank} H_{0,0}(\mathcal{M}) - \sum_{\ell \geq 0} q^\ell \operatorname{rank} H_{1,\ell}(\mathcal{M})$ .                                                     |
| 2501.06662_E00019      | 14   | ■ 0.999 | $\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$ .                                                                                                             | $\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$ .                                                             | $\mu_P(s, u) = \begin{cases} 1 & \text{if } s = u \\ -\sum_{s \leq t < u} \mu_P(s, t) & \text{if } s < u \\ 0 & \text{otherwise} \end{cases}$ .                                                             |
| 2501.06662_E00042      | 19   | ■ 0.999 | $H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1}$ ,                                                                                                                                                                                                               | $H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1}$ ,                                                                                                                                                               | $H(p_x) = -\frac{d}{dt} Z_x(t) \Big _{t=1}$ ,                                                                                                                                                               |
| 2501.06662_E00004      | 5    | ■ 1.000 | $\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$ ,                                                                                                                           | $\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$ ,                                                                           | $\operatorname{Mag}(t\mathcal{M}) = \sum_{\ell} e^{-t\ell} \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$ ,                                                                           |
| 2501.06662_E00012      | 11   | ■ 1.000 | $\begin{aligned} & x = \perp a_1 \cdots a_t \\ & y = \perp a_1 \cdots a_t \cdots a_{t+k} \\ & z = \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$                                                                                      | $\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$                                         | $\begin{aligned} x &= \perp a_1 \cdots a_t \\ y &= \perp a_1 \cdots a_t \cdots a_{t+k} \\ z &= \perp a_1 \cdots a_t \cdots a_{t+k} \cdots a_{t+k+k'} \end{aligned}$                                         |
| 2501.06662_E00031      | 17   | ■ 1.000 | $\prod_{i=1}^k \pi(y_i   y_{i-1})^t = \pi(y_1   y_0)^t \pi(y_2   y_1)^t \pi(y_3   y_2)^t \cdots \pi(y_k   y_{k-1})^t$                                                                                                                                       | $\prod_{i=1}^k \pi(y_i   y_{i-1})^t = \pi(y_1   y_0)^t \pi(y_2   y_1)^t \pi(y_3   y_2)^t \cdots \pi(y_k   y_{k-1})^t$                                                                                       | $\prod_{i=1}^k \pi(y_i   y_{i-1})^t = \pi(y_1   y_0)^t \pi(y_2   y_1)^t \pi(y_3   y_2)^t \cdots \pi(y_k   y_{k-1})^t$                                                                                       |
| 2501.06662_E00007      | 8    | ■ 1.000 | $p(a_3   \perp a) p(a_4   \perp aa_3) p(a_5   \perp aa_3 a_4) p(\dagger   \perp aa_3 a_4 a_5)$ .                                                                                                                                                            | $p(a_3   \perp a) p(a_4   \perp aa_3) p(a_5   \perp aa_3 a_4) p(\dagger   \perp aa_3 a_4 a_5)$ .                                                                                                            | $p(a_3   \perp a) p(a_4   \perp aa_3) p(a_5   \perp aa_3 a_4) p(\dagger   \perp aa_3 a_4 a_5)$ .                                                                                                            |

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| 2501.06662_<br>EQ0008 (1) | 9    | ■ 1.000 | $\pi(y \mid x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                            | $\pi(y \mid x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                            | $\pi(y x) := \left\{ \begin{array}{ll} 1 & \text{if } x = y \\ 0 & \text{if } x \not\rightarrow y \\ \prod_{i=1}^k p(a_{t+i} y_{<t+i}) & \text{if } x \rightarrow y \end{array} \right.$                                                                                                                                                                                                                                                                                                                                                                                                 |
| 2501.06662_<br>EQ0009 (2) | 9    | ■ 1.000 | $\sum_{y \in T(x)} \pi(y \mid x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\sum_{y \in T(x)} \pi(y \mid x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | $\sum_{y \in T(x)} \pi(y x) = \sum_{a \in A \cup \{\dagger\}} p_x(a) = 1,$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| 2501.06662_<br>EQ0010 (4) | 9    | ■ 1.000 | $\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a'a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$ | $\begin{aligned} \sum_{y \in T(x)} \pi(y \mid x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a'a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$ | $\begin{aligned} \sum_{y \in T(x)} \pi(y x) &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a \in A^m} \pi(xa \mid x) \\ &= \sum_{i=0}^{m-1} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{\substack{a=a'a'' \\ a' \in A^{m-1}, a'' \in A}} p(a'' \mid xa') \pi(xa' \mid x) \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \sum_{a'' \in A \cup \{\dagger\}} p(a'' \mid xa') \\ &= \sum_{i=0}^{m-2} \sum_{a' \in A^i} \pi(xa' \dagger \mid x) + \sum_{a' \in A^{m-1}} \pi(xa' \mid x) \end{aligned}$ |
| 2501.06662_EQ0011         | 10   | ■ 1.000 | $\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) &\otimes \mathcal{C}(x, y) \leq \mathcal{C}(x, z) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                              | $\begin{aligned} 1 &\leq \mathcal{C}(x, x) \\ \mathcal{C}(y, z) \otimes \mathcal{C}(x, y) &\leq \mathcal{C}(x, z) \end{aligned}$                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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| 2501.06662_E00013 (6) | 11   | ■ 1.000 | $\begin{aligned} \pi(y \mid x) \pi(z \mid y) &= \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j}) \\ &= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i}) \\ &= \pi(z \mid x). \end{aligned}$                                                                                                                                              | $\pi(y \mid x) \pi(z \mid y) = \prod_{i=1}^k p(a_{t+i} \mid y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} \mid z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} \mid z_{<t+i})$ $= \pi(z \mid x)$                                                                                                                                                                                  | $\pi(y x)\pi(z y) = \prod_{i=1}^k p(a_{t+i} y_{<t+i}) \prod_{j=1}^{k'} p(a_{t+k+j} z_{<t+k+j})$ $= \prod_{i=1}^{k+k'} p(a_{t+i} z_{<t+i})$ $= \pi(z x).$                                                                                                                                                                                                                |
| 2501.06662_E00014     | 12   | ■ 1.000 | $d(x, y) := -\ln \pi(y \mid x).$                                                                                                                                                                                                                                                                                                                                        | $d(x, y) := -\ln \pi(y \mid x).$                                                                                                                                                                                                                                                                                                                                        | $d(x, y) := -\ln \pi(y x).$                                                                                                                                                                                                                                                                                                                                             |
| 2501.06662_E00016     | 13   | ■ 1.000 | $(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$                                                                                                                                                                                                                                                                                                              | $(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$                                                                                                                                                                                                                                                                                                              | $(f * g)(s, u) := \sum_{s \leq t \leq u} f(s, t) g(t, u),$                                                                                                                                                                                                                                                                                                              |
| 2501.06662_E00017     | 13   | ■ 1.000 | $\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$                                                                                                                                                                                                                                                                              | $\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$                                                                                                                                                                                                                                                                              | $\delta(s, u) = \begin{cases} 1 & \text{if } s = u \\ 0 & \text{if } s \neq u \end{cases}$                                                                                                                                                                                                                                                                              |
| 2501.06662_E00018     | 13   | ■ 1.000 | $\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$                                                                                                                                                                                                                                                                                                      | $\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$                                                                                                                                                                                                                                                                                                      | $\zeta_P(s, u) = 1, \quad \text{for all } s \leq u \text{ in } P.$                                                                                                                                                                                                                                                                                                      |
| 2501.06662_E00020     | 14   | ■ 1.000 | $\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$ | $\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$ | $\begin{aligned} \mu_{\mathrm{L}}(x, x) &= 1 \\ \mu_{\mathrm{L}}(x, xa_1) &= -\mu_{\mathrm{L}}(x, x) = -1 \\ \mu_{\mathrm{L}}(x, xa_1a_2) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) = -1 + 1 = 0 \\ \mu_{\mathrm{L}}(x, xa_1a_2a_3) &= -\mu_{\mathrm{L}}(x, x) - \mu_{\mathrm{L}}(x, xa_1) - \mu_{\mathrm{L}}(x, xa_1a_2) = -1 + 1 + 0 = 0. \end{aligned}$ |
| 2501.06662_E00024     | 15   | ■ 1.000 | $\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$                                                                                                                                                                                                                                                                                                            | $\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y \mid x),$                                                                                                                                                                                                                                                                                                            | $\zeta_{\mathcal{M}}(x, y) := e^{-d(x, y)} = \pi(y x),$                                                                                                                                                                                                                                                                                                                 |
| 2501.06662_E00025     | 15   | ■ 1.000 | $\left(\zeta_{\mathcal{M}}\right)_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$                                                                                                                                                                                                                                                                                          | $(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y \mid x)^t.$                                                                                                                                                                                                                                                                                                     | $(\zeta_{\mathcal{M}})_t(x, y) := e^{-td(x, y)} = \pi(y x)^t.$                                                                                                                                                                                                                                                                                                          |
| 2501.06662_E00027 (9) | 16   | ■ 1.000 | $\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$                                                                                                                                                                                                                                                                                                            | $\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k$                                                                                                                                                                                                                                                                                                            | $\zeta_t^{-1} = \sum_{k \geq 0} (-1)^k (\zeta_t - \delta)^k,$                                                                                                                                                                                                                                                                                                           |
| 2501.06662_E00028     | 16   | ■ 1.000 | $(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$                                                                                                                                                                                                                                                                                                     | $(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$                                                                                                                                                                                                                                                                                                     | $(\zeta_t - \delta)(x, y) = \pi(x, y)^t \mathbf{1}_{(x, y) \in E},$                                                                                                                                                                                                                                                                                                     |
| 2501.06662_E00030     | 16   | ■ 1.000 | $\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$                                                                                                                                                                                                                                                                                                   | $\zeta_t^{-1}(x, y) = \pi(y \mid x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$                                                                                                                                                                                                                                                                                                   | $\zeta_t^{-1}(x, y) = \pi(y x)^t \zeta_{\mathrm{L}}^{-1}(x, y).$                                                                                                                                                                                                                                                                                                        |
| 2501.06662_E00035     | 17   | ■ 1.000 | $H_t(p) = \frac{1}{t-1} \left( 1 - \sum_{i=1}^n p_i^t \right).$                                                                                                                                                                                                                                                                                                         | $H_t(p) = \frac{1}{t-1} \left( 1 - \sum_{i=1}^n p_i^t \right).$                                                                                                                                                                                                                                                                                                         | $H_t(p) = \frac{1}{t-1} \left( 1 - \sum_{i=1}^n p_i^t \right).$                                                                                                                                                                                                                                                                                                         |

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| 2501.06662_E00038 | 18   | ■ 1.000 | $\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left( 1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$ | $\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left( 1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$ | $\lim_{t \rightarrow 1} H_t(p) = \lim_{t \rightarrow 1} \frac{1}{t-1} \left( 1 - \sum_{s \in S} p(s)^t \right) = - \sum_{s \in S} p(s) \ln p(s) =: H(p).$ |
| 2501.06662_E00041 | 19   | ■ 1.000 | $Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t \ln p_x(a)}$                                                   | $Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t \ln p_x(a)}$                                                   | $Z_x(t) = \sum_{a \in A \cup \{\dagger\}} p_x(a)^t = \sum_{a \in A \cup \{\dagger\}} e^{-t \ln p_x(a)}$                                                   |
| 2501.06662_E00043 | 19   | ■ 1.000 | $\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$                 | $\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$                 | $\mathbb{E}_\rho(E) = \sum_{s \in S} E(s) \frac{e^{-tE(s)}}{\tilde{Z}(t)} = -\frac{d}{dt} \ln \tilde{Z}(t) = \frac{f'(t)}{\tilde{Z}(t)},$                 |
| 2501.06662_E00047 | 20   | ■ 1.000 | $G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$      | $G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$      | $G_{k,\ell} = \left\{ (y_0, \dots, y_k) \in M^{k+1} \mid \sum_{i=0}^{k-1} d(y_i, y_{i+1}) = \ell \text{ and for all } i, y_i \neq y_{i+1} \right\}.$      |
| 2501.06662_E00048 | 20   | ■ 1.000 | $\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$                                                                                                           | $\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$                                                                                                           | $\partial_k = \sum_{i=0}^k (-1)^i \partial_k^i$                                                                                                           |
| 2501.06662_E00050 | 21   | ■ 1.000 | $\operatorname{Mag}(t, \mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$                             | $\operatorname{Mag}(t, \mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$                             | $\operatorname{Mag}(t, \mathcal{M}) = \sum_{\ell} q^\ell \sum_{k \geq 0} (-1)^k \operatorname{rank}(H_{k,\ell}(\mathcal{M}))$                             |
| 2501.06662_E00053 | 21   | ■ 1.000 | $\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \# G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$                  | $\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \# G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$                  | $\sum_{c \in G_{k,\ell}} (-1)^k e^{-t\ell} = (-1)^k q^\ell \# G_{k,\ell} = (-1)^k q^\ell \operatorname{rank}(MC_{k,\ell}(\mathcal{M})).$                  |
| 2501.06662_E00055 | 22   | ■ 1.000 | $PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i   y_{<i}) \right\}$                                                                            | $PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i   y_{<i}) \right\}$                                                                            | $PPL(x) = \exp \left\{ -\frac{1}{n} \sum_{i=1}^n \ln p(a_i   y_{<i}) \right\}$                                                                            |
| 2501.06662_E00057 | 23   | ■ 1.000 | $\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$                                                                                  | $\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$                                                                                  | $\theta(a, b) = 0 \quad \text{if} \quad \mathrm{A}(a, b) = \varnothing.$                                                                                  |
| 2501.06662_E00060 | 23   | ■ 1.000 | $-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$                                                                                    | $-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$                                                                                    | $-\sum_{a \in A} \sum_{a' \in A} p_{\perp}(a) P(a, a') \log P(a, a'),$                                                                                    |